Contact

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www.linkedin.com/in/m-tanner (LinkedIn)
friendlyventures.io (Personal)
github.com/m-tanner (Portfolio)

Top Skills

Agentic AI Development

Large Language Models (LLM)

Python

Languages

English (Native or Bilingual)

German (Professional Working)

Chinese (Elementary)

Honors-Awards

Rod and Connie Radovanovic Scholarship

Space Shuttle Scholarship

Eagle Scout Scholarship

Michael Tanner

Senior Software Engineer at Axon

Greater Seattle Area

Summary

Ex-SpaceX, ex-Blue Origin. I like ambitious projects that need someone to define the product, draw the architecture, write the code, and lead the team. Most recently shipped Axon Community Request with Amazon Ring as tech lead, taking the project from greenfield design through public launch over five years.

Experience

Axon

4 years 2 months

Senior Software Engineer

July 2024 - Present (2 years 1 month)

Seattle, WA

• Delivered Axon Community Request with Ring as technical lead of a 5-engineer team the publicly announced Axon–Ring partnership (axon.com/products/community-request/ring) that lets verified public safety agencies invite Ring camera owners to securely share video evidence, with full chain-of-custody preserved into Axon Evidence

• Resurrected and shipped a previously paused greenfield project and rolled it out to Ring Neighbors-Verified agencies across the US

• Owned system design, sprint execution, and code review across the team while interfacing with product, legal, and partner engineering & security counterparts to deliver a privacy-preserving, auditable, and courtroom-ready evidence pipeline

• Led the team to build AI-Powered Crash Diagramming during Axon's company-wide Gen-AI hackathon with the prototype selected by Axon's CEO roundtable and voted a top-3 favorite by customers at IACP

• Built and shipped the NCIC Store backend for Axon's records-management product. Within weeks of launch, 103 Ft. Lauderdale PD officers had completed 600+ imports (~45/day), with five additional Florida agencies through UAT and Axon's professional services team replicating the rollout to new agencies without engineering involvement.

Software Engineer II

June 2022 - July 2024 (2 years 2 months)

Seattle, WA

• Co-owned search and indexing for Axon Evidence (evidence.com), the digital evidence management system used by public safety agencies worldwide, across 9 production Solr clusters spanning US, EU, and government/regulated environments

• Migrated production search infrastructure to Solr 8 / Java 11 with remote-collection joins, including personal rollout of 3/9 clusters (including the largest one) and rebuilt the migration tooling to handle simultaneous version-and-topology changes that the prior scripts couldn't

• Shipped Lucene's unified highlighter to production and migrated transcript search, cutting per-query latency ~4× (~2s to ~0.5s) for a 15% index-size increase, with no regressions during migration

• Led after-action reviews for production incidents, partnering with SRE, infrastructure, and engineering executives on root cause, and closed action items spanning dashboards, infra, and code

DevZero

Founding Engineer

January 2022 - June 2022 (6 months)

Seattle, WA

• Founding technical staff at a Seattle-based startup building ephemeral, ready-to-code, on-demand cloud developer environments

• Designed and shipped end-to-end features in Go across the platform during the company's earliest months

Axon

Software Engineer II

September 2020 - January 2022 (1 year 5 months)

Seattle, WA

• Lead engineer on the greenfield design of what shipped publicly in 2025 as Axon Community Request with Ring. Owned product definition, system architecture, and detailed design for a new way for police to collect digital evidence from communities

• Defined the chain-of-custody model, geo-targeted request flow, and voluntary-share/anonymous-decline pattern that became the foundation of the shipped product integrating Ring's Neighbors feed with Axon Evidence

• Partnered with product, legal, and security stakeholders to scope a system that handles community-sourced digital evidence at scale while preserving resident privacy and auditability

BLUE ORIGIN

2 years 11 months

Software Engineer

March 2019 - August 2020 (1 year 6 months)

Seattle, WA

• Cut away 6,000 engineering-hours/year by building a web service for time-series analysis by employing a Python and MongoDB back-end with a Flask and Jinja2 front-end, hosted on Kubernetes with testing/deployment via GitLab CI

• Further reduced data processing time from 210 to 20 minutes for > 1,000 test sessions by leveraging multiprocessing, enabling fast batch processing

• Slashed cost of employees searching for engineering drawings by 80% by recognizing customer needs, iterating on, and rapidly releasing a product using Terraform, Gitlab CI, Python, AWS Lambda, API Gateway, and Kubernetes

• Minimized cost of business intelligence extract endpoints by 99% by using Terraform, Gitlab CI, Java, Spring Boot, AWS Lambda and API Gateway

• Halved the time required for post-test visual inspections by developing Python to operate a custom-built robotic inspection tool

Propulsion Engineer

October 2017 - March 2019 (1 year 6 months)

• Supported a rocket engine ignition system, including design, modeling, analysis, drafting, manufacturing by metallic 3D-printing, testing, and data review

• Developed a valve manifold for the BE-4 ignition system, including CAD modeling in Creo, drawing in Creo using GD&T, internal release processes via Windchill, manufacturing by exotic metal 3D-printing and traditional machining

• Intro'd semi-automated inspection tracking with Jira, which improved efficiency and allowed for the first-ever two Blue Engine 4 (BE-4) tests in a single day

• Assembled, inspected, troubleshot, and conducted tests on the most powerful liquid oxygen/liquid methane rocket engine to date, BE-4

SpaceX
Associate Launch Engineer
May 2017 - August 2017 (4 months)
Cape Canaveral, FL

• Predicted, using heat transfer concepts, Excel, and VBA, and analyzed inter-stage purge heat exchanger performance across launch sites

• Developed VBA code to plan, procure, and install secondary cooling supply water for helium hydropacs

Maurice J. Zucrow Laboratories
Undergraduate Research Assistant
April 2016 - April 2017 (1 year 1 month)
Purdue University

• Designed, fabricated, and analyzed a sub-scale, bi-propellant test stand to meet needs for testing a hypergolic H_2O_2/hydrocarbon catalyst combustor

• Predicted flow rates, using Excel, REFPROP, and VALCOR, for fluid system sizing and fabrication

• Built the test stand, including tube bending and flaring, while meeting requirements for oxidizer cleanliness

• Handled dangerous and confidential propellants properly while disassembling a previous test rig to reuse components

• Conducted an uncertainty analysis on results using MATLAB and presented results in a final paper

SpaceX

Manufacturing Engineering Intern

May 2016 - August 2016 (4 months)

Hawthorne, CA

• Trained on builds, created part numbers, authored engineering masters, attained approvals, worked closely with techs on first builds, redlined, and dispositioned issues for the integration of final assembly bench work at component level of the bi- propellant throttle valve (BTV), BTV actuator, and liquid oxygen pre-valves, reducing test stand downtime in one case by 5 days

• Communicated and coordinated extensively between avionics, engine assembly, components assembly, analysis, and design groups

• Worked closely with technicians to design in NX, 3D-print, and iterate nesting tools for problematic builds

SpaceX

Ground Support Equipment Engineering Intern

September 2015 - December 2015 (4 months)

McGregor, TX

• Designed using VALCOR and VISIO, fabricated, installed, and activated a warming purge system to decrease post-test downtime on Merlin Stand, including lock-out tag-out, hands-on high pressure plumbing installation, purges, and leak checks

• Designed using NX, fabricated, installed, and performed extensive troubleshooting to allow for better filtration and density/temp. data acquisition on an upgraded fuel offloading system

• Designed using NX, communicated closely with a vendor to fabricate, and installed protection for vacuum jacketed piping and test stands

Daimler AG

Body Shop Planning Intern

January 2015 - May 2015 (5 months)

Sindelfingen, Germany

• Collected data for takt time simulations which determined and helped plan necessary buffer sizes for several body shop stations

• Influenced shop design in workshops to determine cell layout, process flow, and product flow considering a new body shop and product line

Mercedes-Benz USA
Paint Process Engineering Intern
May 2014 - August 2014 (4 months)
Vance, AL

• Analyzed all measurement gauges statistically using MiniTab

• Developed VBA for Excel code to compile, analyze, and present shop downtime
data, which enabled leadership to address largest causes

Education

Purdue University
Bachelor of Science - BS, Mechanical Engineering, Global Engineering · (2012 - 2017)

Bellevue College
Computer Science · (2019 - 2020)

Karlsruhe Institute of Technology (KIT)
Mechanical Engineering · (2015 - 2015)